

Controlled Spatial-Convergence Trio — Run Results

Auto-generated from run data; companion to report_v6_full

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GREEN – Rev.6 REPRODUCED: comparability + plateau gates pass, controlled trio significantly non-monotone ($S_{\text{rev}}=874$), all values within 2% of archival

Run metadata

Generated 2026-08-21 20:50:59; code commit 676e74f (dirty). Nodes: data/she2d_richardson_{71x58,100x70,141x85}

Comparability gates

mesh-law/solver signatures identical	YES
frozen H -array SHA1 identical	YES (b4319bb03303)
drain- dx refinement ratios	1.41436, 1.41437 (target $\sqrt{2} = 1.41421$)
observable-plateau gate ($\leq 3e-04$, both G_{ii} & G_{tot})	PASS

Frozen trio (this run)

node	drain dx (μm)	$G_{ii,\text{max}}$ ($\text{cm}^{-3}\text{s}^{-1}$)	spread (G_{ii})	G_{tot} ($\text{cm}^{-1}\text{s}^{-1}$)	spread (G_{tot})	R_ϕ (V)
G_1 (71×58)	0.0038	2.990×10^{25}	$1.4e-04$	8.982×10^{13}	$2.3e-04$	$8.7e-05$
G_2 (100×70)	0.0027	2.135×10^{25}	$2.9e-04$	6.027×10^{13}	$2.2e-04$	$5.8e-04$
G_3 (141×85)	0.0019	2.413×10^{25}	$6.1e-05$	6.593×10^{13}	$5.2e-05$	$1.8e-04$

spread = $(\max_4 - \min_4)/\bar{G}_4$; plateau band $u = (\max_4 - \min_4)/2$.

Verdict and classification

$G_{ii,\text{max}}$: $2.990 \rightarrow 2.135 \rightarrow 2.413 \times 10^{25}$; $\Delta_{12} = +8.555e + 24$, $\Delta_{23} = -2.781e + 24$ — NON-MONOTONE, $S_{\text{rev}} = 874 \gg 1$.

Classification: GREEN.

Rule. **GREEN**: all comparability+plateau gates pass, sequence significantly non-monotone ($S_{\text{rev}} > 3$), and every G_i within 2% of archival ($2.990/2.135/2.413 \times 10^{25}$). **YELLOW**: all scientific gates pass and still significantly non-monotone, but ≥ 1 value outside that tolerance. **RED**: verdict not reproduced or run invalid — monotone sequence, unresolved reversal ($S_{\text{rev}} \leq 3$), H -hash/signature mismatch, or a node failing the plateau gate.

Timing / convergence

node	wall / iters / s per iter	final R_ϕ (V)	$T_{e,\max}$ (K)
71×58	n/a	$8.7e-05$	3177
100×70	n/a	$5.8e-04$	3161
141×85	n/a	$1.8e-04$	3196

See `data/richardson_trio_run.log` and `richardson_trio_metrics.csv` for the full per-iteration history and any warnings. Scientific interpretation: `report_v6_full`, § Controlled spatial-convergence.